Computational Physics Classes No.6 - Report

Simulation of heat diffusion with Crank-Nicolson method

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Abstract

This report presents a numerical study of two–dimensional heat diffusion in a square room using the Crank–Nicolson method. The model incorporates mixed thermal boundary conditions, a localized heater with thermostat control, and a window region with enhanced heat loss. The temperature field is evolved using an implicit finite–difference scheme solved by Gauss–Seidel relaxation, and the energy supplied and lost through the boundary is tracked at each time step. Several physical scenarios are examined, including insulated walls, weak and strong heat leakage, regulated heating, and extreme external temperatures. The results demonstrate the stability and accuracy of the method, as well as the strong influence of boundary conditions on thermal behaviour.

1 Introduction

Heat diffusion is one of the fundamental processes studied in physics and engineering, describing how thermal energy spreads within a material due to temperature gradients. Analytical solutions exist only for simple geometries and boundary conditions; therefore, numerical methods play a crucial role in solving realistic heat–transfer problems. In this report, we investigate a two–dimensional model of heat diffusion inside a square room with mixed boundary conditions, internal heat sources, and a controllable heating system.

The goal of this project is to implement and analyze a numerical simulation based on the Crank–Nicolson method, which is an implicit, second–order accurate scheme known for its stability and efficiency. The simulation incorporates a window with enhanced heat loss, a heater operating according to a feedback control rule, and the ability to compute energy supplied to and lost from the system. Using this computational model, we explore several scenarios that illustrate the thermal behaviour of the room under different physical conditions.

The results presented here include the temporal evolution of the temperature at a specific point, spatial temperature distributions at selected times, and energy–exchange characteristics. These findings demonstrate the capability of the Crank–Nicolson scheme to model realistic heat–transfer phenomena and provide insight into how boundary conditions and heating strategies affect the thermal dynamics of confined spaces.

2 Description of the Numerical Method

The temperature evolution inside the room is governed by the two–dimensional heat–diffusion equation with an internal heat source:

$$\frac{\partial T(x,y,t)}{\partial t} = D\nabla^2 T(x,y,t) + S(x,y,t), \tag{1}$$

where D is the thermal diffusivity and $S(x,y,t)$ represents the power density of the heater. The computational domain is a square of size $L \times L$, discretized into an $N \times N$ grid with spatial step $\Delta = L/(N-1)$. Time is discretized using a constant time step Δt.

Crank–Nicolson Scheme

To evolve the temperature field in time, we apply the Crank–Nicolson method, which is an implicit finite–difference scheme with second–order accuracy both in time and space. Applying the method to the diffusion equation leads to the following discretized form:

$$T_{i,j}^{n+1} - \alpha\left(T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1}\right) = R_{i,j}^{n}, \tag{2}$$

where

$$\alpha = \frac{D\Delta t}{2\Delta^2}, \tag{3}$$

and $R_{i,j}^n$ is the right–hand side term incorporating the known temperatures from the previous time step:

$$R_{i,j}^{n} = T_{i,j}^{n} + \alpha\left(T_{i+1,j}^{n} + T_{i-1,j}^{n} + T_{i,j+1}^{n} + T_{i,j-1}^{n} - 4T_{i,j}^{n}\right) + \frac{\Delta t}{2} S_{i,j}^{n}. \tag{4}$$

Gauss–Seidel Relaxation

Since the Crank–Nicolson scheme produces a system of linear equations at every time step, an iterative solver is required. We employ the Gauss–Seidel relaxation method, which updates the temperature at each interior node according to

$$T_{i,j}^{n+1} \leftarrow \frac{R_{i,j}^{n} + \alpha\left(T_{i-1,j}^{n+1} + T_{i+1,j}^{n+1} + T_{i,j-1}^{n+1} + T_{i,j+1}^{n+1}\right) + \frac{\Delta t}{2} S_{i,j}^{n+1}}{1 + 4\alpha}. \tag{5}$$

Iterations continue until the residual of the discretized equation falls below a chosen tolerance.

Boundary Conditions

Three types of thermal boundaries are included in the model:

- **Insulated walls:** implemented as Neumann-type boundaries ($\partial T/\partial n = 0$), obtained by setting $h = 0$ in the Robin condition.
- **Walls with heat exchange:** modeled using the Robin condition

  $$-D\frac{\partial T}{\partial n} = h\left(T - T_{\infty}\right), \tag{6}$$

  where h is the heat transfer coefficient.
- **Window region:** a section of the eastern wall with an enhanced heat–loss coefficient $h_w > h$.

These boundary conditions are incorporated into the iterative solver at every relaxation step.

Heater Model

The heat source is localized in a rectangular region inside the room and operates with power density $S_{\max}$. The heater is switched on or off depending on the temperature at a control point (i_c, j_c):

$$w = 1, \quad \text{if } T(i_c, j_c) < T_{\text{low}}, \tag{7}$$

$$w = 0, \quad \text{if } T(i_c, j_c) > T_{\text{high}}, \tag{8}$$

with linear interpolation of the source term between time steps according to the Crank–Nicolson scheme.

Energy Accounting

To analyze heat gain and loss, two energy measures are computed at each time step:

- Energy supplied by the heater:

$$E^n_{\text{sup}} = \frac{1}{2}(w^n + w^{n-1})\Delta^2 \Delta t \sum_{i,j} S_{i,j}, \tag{9}$$

- Energy lost through the window:

$$E^n_{\text{win}} = \Delta \Delta t \sum_{j=j_1}^{j_2} h_w \left(T^n_{N-1,j} - T_\infty\right), \tag{10}$$

where j_1 and j_2 denote the indices of the window segment. These quantities enable examination of energy balance in different physical scenarios.

Overall, the combination of the Crank–Nicolson scheme, Gauss–Seidel relaxation, and physically accurate boundary conditions provides a stable and efficient method for solving the heat–diffusion problem under various configurations.

3 Results

This section presents numerical simulations for the six physical scenarios specified in the project description. Following the convention used during the laboratory classes, the results for Tasks 1–4 are arranged into four vertical columns: each column begins with the time evolution of the temperature at the control point (i_c, j_c), followed by four spatial temperature maps corresponding to selected time instants. For Task 4, the snapshots were taken at the specific times marked with full dots in the sensor temperature plot.

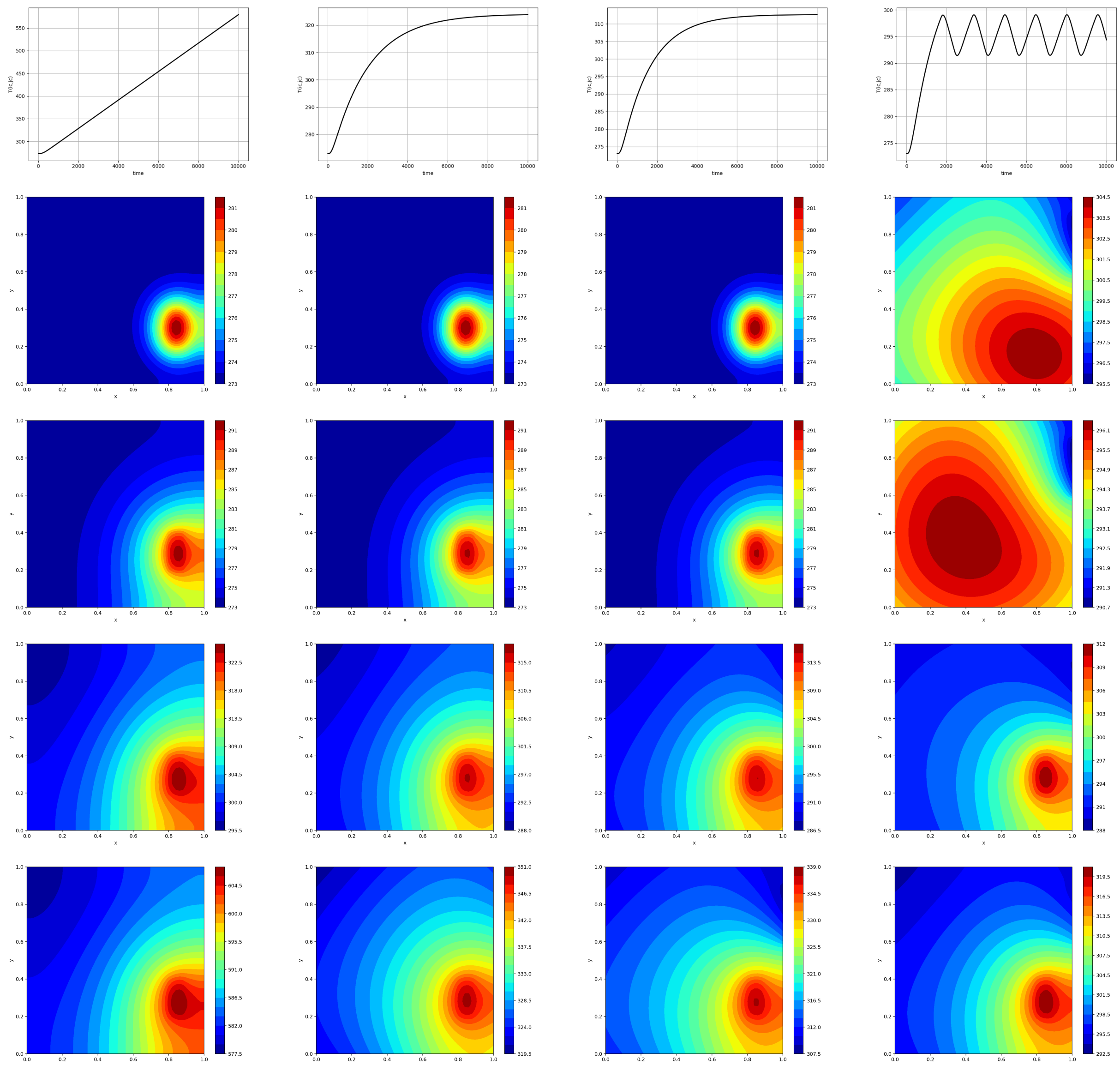

Figure 1: Tasks 1–4. Each column contains: (top) time variation of the temperature at the control point, (below) four spatial temperature distributions at selected time instants. In Task 4 (fourth column), the shown snapshots correspond to times marked with full dots in the top plot.

3.1 Broken Window

In Task 5 the window heat transfer coefficient is increased to $h_w = 1.0$, causing strong heat loss to the exterior. The heater must therefore supply significantly more energy to maintain the temperature near the required threshold.

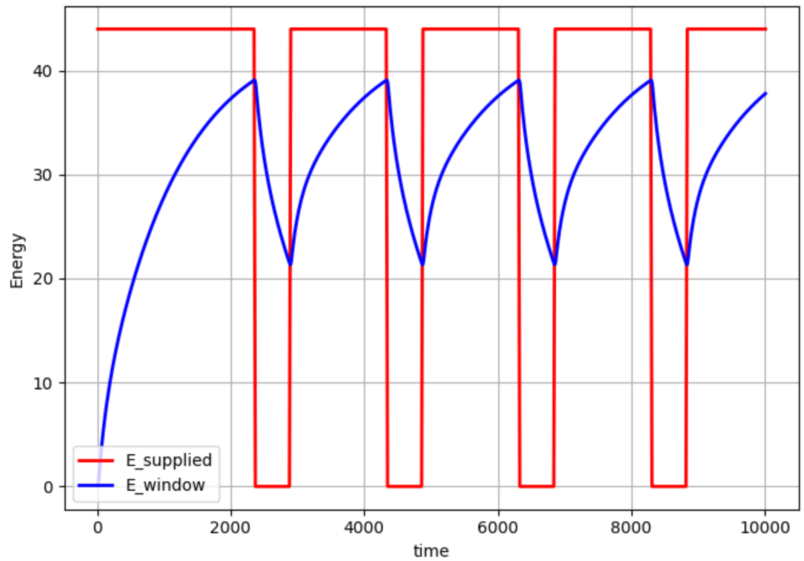

Figure 2: Energy supplied by the heater and energy lost through the window for Task 5.

3.2 Extremely Hot Exterior

In this scenario, the external temperature is set to an extreme value $T_\infty = 10^4\,\mathrm{K}$, resulting in rapid heating of the room interior. The energy balance is shown in Figure 3.

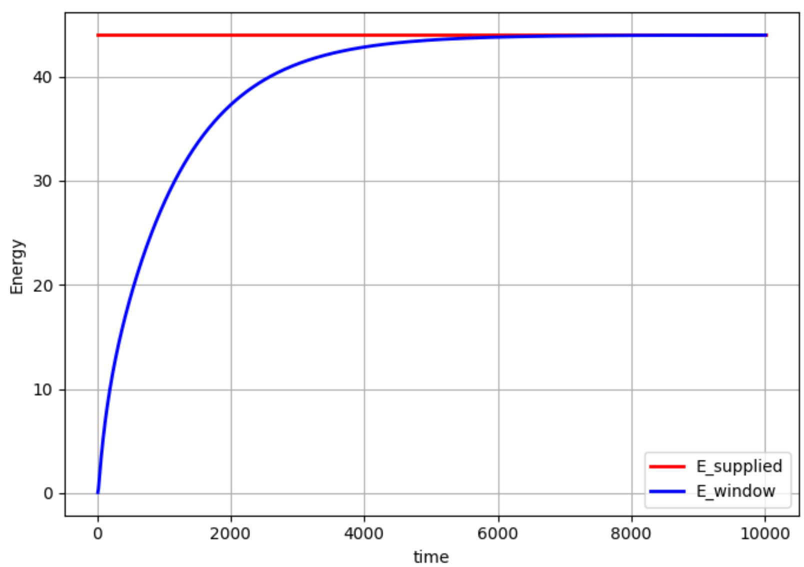

Figure 3: Energy exchange in Task 6: heat gained from a very hot exterior.

4 Conclusions

In this project, we implemented a two–dimensional numerical model of heat diffusion using the Crank–Nicolson method combined with Gauss–Seidel relaxation and realistic thermal boundary conditions. The simulation framework allowed us to investigate how different physical settings—such as wall insulation, window heat loss, and thermostat control—modify the temperature distribution and energy balance inside the modeled room.

The results demonstrate the stability and effectiveness of the Crank–Nicolson scheme in handling diffusion processes with mixed boundary conditions. For fully insulated walls (Task 1), the temperature increases monotonically due to the absence of heat loss. Introducing even a small heat–transfer coefficient (Task 2) leads to a well-defined equilibrium temperature, while enhancing heat loss through the window (Task 3) produces a distinct spatial asymmetry in the temperature field. The thermostat mechanism investigated in Task 4 shows that on/off switching can successfully maintain the temperature near a desired threshold, although with characteristic oscillations related to the heating cycle.

Energy analysis in Tasks 5 and 6 highlights the significant impact of the external environment. A broken window dramatically increases the energy required from the heater, while an extremely hot exterior reverses the heat flow and rapidly warms the interior of the room. These scenarios illustrate the importance of boundary conditions in thermal engineering applications.

Overall, the implemented solver provides accurate, stable, and physically meaningful results across all tested configurations. The study confirms that the Crank–Nicolson method, combined with iterative solvers and appropriate physical modeling, is a powerful tool for simulating heat diffusion in realistic geometries.

References

[1] Dr. Hab. Eng. Chwiej, T. *Project: Simulation of heat diffusion with Crank-Nicolson method*. AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/6_heat_diffusion.pdf